

**MATHEMATICS CLASS – X WBBSE 2019**

1. Choose the correct option in each case from the following questions: (1X6=6)

- i. In a partnership business, the ratio of share of profit of two friends is  $\frac{1}{2}:\frac{1}{3}$ , then the ratio of their principal is \_\_\_\_\_  
(a) 2:3 (b) 3:2 (c) 1:1 (d) 5:3
- ii. If  $p + q = \sqrt{13}$  and  $p - q = \sqrt{5}$  then the value of  $pq$  is \_\_\_\_\_  
(a) 2 (b) 18 (c) 9 (d) 8
- iii. O is the centre of the circle and AB a diameter. ABCD is a cyclic quadrilateral  $\angle ABC = 65^\circ$ ,  $\angle DAC = 40^\circ$ , then the measure of  $\angle BCD$  is \_\_\_\_\_  
(a)  $75^\circ$  (b)  $105^\circ$  (c)  $115^\circ$  (d)  $80^\circ$
- iv. If  $\tan \alpha + \cot \alpha = 2$ , then value of  $\tan^{13} \alpha + \cot^{13} \alpha =$  \_\_\_\_\_  
(a) 13 (b) 2 (c) 1 (d) 0
- v. If two cubes of length of each side  $2\sqrt{6}$  cm are placed side by side, then the length of the diagonal of the cuboid so produced is \_\_\_\_\_  
(a) 10 cm (b) 6 cm (c) 2 cm (d) 12 cm.
- vi. The mean of the data  $x_1, x_2, x_3, \dots, x_{10}$  is 20 then the mean of  $x_1 + 4, x_2 + 4, x_3 + 4, \dots, x_{10} + 4$  will be \_\_\_\_\_.  
(a) 20 (b) 24 (c) 40 (d) 10

2. Fill in the blanks (any Five):- (1X5=5)

- i. A person deposited Rs. 100 in a bank and gets the amount Rs. 121 after two years. The rate of compound interest is \_\_\_\_\_%.
- ii. If the product and sum of two quadratic surds is a rational number, then surds are \_\_\_\_\_ surds.
- iii. If the bases of two triangles are situated on same line and the other vertex of two triangles are common, then the ratio of areas of two triangles are \_\_\_\_\_ to the ratio of their bases.
- iv. The simplest value of  $\frac{\cos 53^\circ}{\sin 37^\circ}$  is \_\_\_\_\_.
- v. Number of surfaces of a solid right circular cylinder is \_\_\_\_\_.
- vi. The variables  $x_1, x_2, x_3, \dots, x_{100}$  are in ascending order of their magnitude, then the median of the variable is \_\_\_\_\_.

3. Write True or False (any Five):- (1X5=5)

- i. The difference between the simple interest and the compound interest of Rs. 100 in 1 year at the rate of 10% p.a. is Re. 1.
- ii. The compound ratio of  $ab:c^2, bc:a^2, ca:b^2$  is 1:1.
- iii. Only one circle can be drawn through three non-collinear points.
- iv.  $\sin 30^\circ + \sin 60^\circ > \sin 90^\circ$ .

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- v. The ratio of the volume of a right circular cone and a right circular cylinder with same base and height 1:3.
- vi. Value of median of data 2, 3, 9, 10, 9, 3, 9 is 10.
4. Answer the following questions (any ten):- (2X10=20)
- i. Find the capital which gives Re. 1 as interest per month at 5% rate of interest per annum.
- ii. In a partnership business the ratio of capitals of three men is 3:5:8. The share of profit of the first member is Rs. 60 less than that of the third member, then what is the total profit in the business?
- iii. If  $\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{2a-3b+4c}{p}$ , then find p.
- iv.  $x \propto y^2$  and  $y = 2a$  when  $x = a$ , then show that  $y^2 = 4ax$ .
- v. In a trapezium ABCD,  $BC \parallel AD$  and  $AD = 4$  cm. The two diagonals AC and BD intersect at the point O in such a way that  $\frac{AO}{OC} = \frac{DO}{OB} = \frac{1}{2}$ . Calculate the length of BC.
- vi. Two chords AB and AC of a circle mutually perpendicular to each other. If  $AB = 4$  cm and  $AC = 3$  cm, find the length of the radius of the circle.
- vii. In triangle ABC,  $\angle ABC = 90^\circ$ ,  $BD \perp AC$ . If  $AB = 5$  cm,  $BC = 12$  cm, then find the length of BD.
- viii. Find the value/values of  $\theta$  ( $0^\circ \leq \theta \leq 90^\circ$ ) for which  $2 \sin \theta \cos \theta = \cos \theta$ .
- ix. If  $\sin 10\theta = \cos 8\theta$  and  $10\theta$  is a positive acute angle; then find the value of  $\tan 9\theta$ .
- x. If the length, breadth and height of a cuboidal room be a unit, b unit and c unit respectively and  $a + b + c = 25$ ,  $ab + bc + ca = 240.5$ , then find the length of the longest rod to be kept inside the room.
- xi. The area of the curved surface of a right circular cone is  $\sqrt{5}$  times that of the base of the cone, find the ratio of the height and the radius of the base of the cone.
- xii. The mid value of the first  $(2n+1)$  consecutive natural numbers is  $\frac{n+103}{3}$ .  
Find the value of  $n$ .
5. Answer any one question:- (1X5=5)
- i. If the interest is compounded half yearly what will be the compound interest and amount on Rs. 8000/- at the rate of 10% compound interest per annum for  $1\frac{1}{2}$  years?
- ii. Two friends start a partnership business investing Rs. 40000 and Rs. 50000 respectively. There is an agreement between them that 50% of the profit will be divided equally and rest amount of profit will be distributed

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between them in the ratio of their principal. If the share of profit of 1<sup>st</sup> friend is Rs. 800 less than that of 2<sup>nd</sup> friend, find out the share of profit of the 1<sup>st</sup> friend.

6. Solve any one question:-

(1X3=3)

- Determine the equation whose roots are the square of the roots of the equation  $x^2 + x + 1 = 0$ .
- If the price of 1 dozen pens is reduced by Rs. 6 then 3 more pens will be got in Rs. 30. Calculate the price of 1 dozen pen before the reduction of price.

7. Answer any one question:-

(1X3=3)

- Simplify  $\frac{4\sqrt{3}}{2-\sqrt{2}} - \frac{30}{4\sqrt{3}-\sqrt{18}} - \frac{\sqrt{18}}{3-\sqrt{12}}$ .
- If  $\left(\frac{1}{x} - \frac{1}{y}\right) \propto \frac{1}{x-y}$ , then show that  $(x^2 + y^2) \propto xy$

8. Answer any one question:-

(1X3=3)

- If  $(3x - 2y) : (x + 3y) = 5 : 6$ , then find the value of  $(2x + 5y) : (3x + 4y)$ .
- If  $\frac{b+c-a}{y+z-x} = \frac{c+a-b}{z+x-y} = \frac{a+b-c}{x+y-z}$ , then prove that  $\frac{a}{x} = \frac{b}{y} = \frac{c}{z}$ .

9. Answer any one question:-

(1X5=5)

- Prove that semicircular angle is a right angle.
- If two circles touch each other externally then the point of contact will lie on the line segment joining the two centres – prove it.

10. Answer any one question:-

(1X3=3)

- If a quadrilateral ABCD is circumscribed about a circle with centre O, prove that  $AB + CD = BC + DA$ .
- If in a triangle ABC,  $\angle A$  is right angle and BP and CQ are two medians, then prove that  $5BC^2 = 4(BP^2 + CQ^2)$ .

11. Answer any one question:-

(1X5=5)

- Draw a triangle ABC of which  $BC = 7$  cm,  $AB = 5$  cm and  $AC = 6$  cm. Then draw the circumcircle of triangle ABC (only traces of construction required)
- Construct a circle of radius 4 cm and draw two tangents to the circle from an external point at a distance 6.5 cm from the centre of the circle.

12. Answer any two questions:-

(3X2=6)

- In triangle ABC,  $\angle C = 90^\circ$ , if  $BC = m$  and  $AC = n$ , then prove that  $m \sin A + n \sin B = \sqrt{m^2 + n^2}$ .
- Find the value of  $\frac{4}{3} \cot^2 30^\circ + 3 \sin^2 60^\circ - 2 \operatorname{cosec}^2 60^\circ - \frac{3}{4} \tan^2 30^\circ$ .
- If  $\angle P + \angle Q = 90^\circ$ , then show that  $\sqrt{\frac{\sin P}{\cos Q}} - \sin P \cos Q = \cos^2 P$ .

**MATHEMATICS CLASS – X WBBSE 2019****13. Answer any one question:-****(1X5=5)**

- i. From a quay of a river, 600 metres wide, two boats in two different directions to reach the opposite side of the river. The first boat moves making an angle  $30^\circ$  with this bank and second boat moves making an angle  $90^\circ$  with the direction of first boat. What will be the distance between the two boats when both of them reach other side?
- ii. The length of a flag post at the roof of a three storied building is 3.6 metre. The angles of elevation of the top and foot of the post are  $50^\circ$  and  $45^\circ$  respectively from a point on the road. Find the height of the building. [Take  $\tan 50^\circ = 1.2$ ]

**14. Answer any two questions:-****(4X2=8)**

- i. If 64 buckets of water is withdrawn from a cubical water tank, full of water, then  $\frac{1}{3}$  of water in the tank remains. If the length of the side of the water tank is 1.2 metre then what is the capacity of the each bucket (in litre)? (1 cubic decimeter = 1 litre)
- ii. The diameter of cross section of a wire is reduced by 50%. If the volume remains constant, what percent of length of the wire should be increased?
- iii. 77 sq. m tripal is required to make a right circular conical tent. If the slant height of the tent is 7 m, then what is the area of the base of the tent?

**15. Answer any two questions:-****(4X2=8)**

- i. If the arithmetic mean of the following frequency distribution tale is 54, then find the value of K:

| Class     | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 |
|-----------|------|-------|-------|-------|--------|
| Frequency | 7    | 11    | K     | 9     | 13     |

- ii. Making frequency distribution table from the given cumulative frequency distribution table, find the mode of the data:

| Class                | Less Than 10 | Less Than 20 | Less Than 30 | Less Than 40 | Less Than 50 | Less Than 60 | Less Than 70 | Less Than 80 |
|----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Cumulative Frequency | 4            | 16           | 40           | 76           | 96           | 112          | 120          | 125          |

- iii. Find the average marks of 52 students of class X in a school by using direct method and assumed mean method from the table given below:

| No. of Students | 4  | 7  | 10 | 15 | 8  | 5  | 3  |
|-----------------|----|----|----|----|----|----|----|
| Marks           | 30 | 33 | 35 | 40 | 43 | 45 | 48 |

**[Additional Questions for External Candidates]**

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**16. (a) Answer any three questions:**

**(2X3=6)**

- i. If  $p:q=5:7$  and  $p+q=-4$  then what is the value of  $(3p+2q)$ ?**
- ii. In how many years the interest will be  $\frac{3}{5}$ th of principal at the rate of 10% simple interest per annum?**
- iii. What is the circular value of an angle formed by the end of the point of hour hand of a clock in 1 hr rotation?**
- iv. If  $x=2+\sqrt{5}$  and  $xy=-1$ , find the value of  $x-y$ .**

**(b) Answer any four questions:-**

**(1X4=4)**

- i. What is the value of  $a$  if one of the root of the equation  $x^2+ax+3=0$  is 1?**
- ii. If the product of three positive continued proportional numbers is 64, what is their mean proportional?**
- iii. Find the value of  $k$  if the roots of the equation  $x^2-kx+4=0$  are real and equal.**
- iv. Find the median of the numbers 1, 2, 3, 5, 8, 6, 9, 11 and 4.**
- v. In a partnership business A invests Rs. 600 for 9 months and B invests Rs. 700 for 5 months, find the ratio of their share of profit.**

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